



What Must Come First?

Some steps depend on other steps

Name:

Date:

★ I can find the step that must come before another step.

Steps that depend on steps

In an algorithm, some steps **MUST** come before others. A step that needs another step has to come after it. Some steps can go in either order. Read the tea algorithm and answer the questions.

Make a cup of tea — the steps

A. add the milk

B. boil the water

C. put a tea bag in the cup

D. pour the water in the cup

1 What must come first?

You can only pour the water (D) once it is hot. Which step makes the water hot and must come before D? Write the letter.

2 Can it go early?

Can you add the milk (A) before you pour the water (D)? Circle YES or NO and tell why.

3 You try — order it

Two steps can be done in either order. Which two letters? Then write the whole tea algorithm in an order that works.

Big idea

When one step depends on another, it must come after it. Some steps don't depend on each other, so they can go in any order. Finding what must come first helps you plan a good algorithm.



What Must Come First?

Quick facilitation notes & answer key

What this worksheet teaches

Students reason about which step must come before another, which cannot go early, and which two steps are flexible — then write a working order for making tea. Seeing that some steps depend on others is key to a correct plan. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: cut the steps into cards so students can physically slide them into order before writing.

How to lead it (about 20 minutes)

1. Show them (you do). You can only pour the water once it is hot — so boiling must come before pouring.
2. Do one together. Find the step that must come before pouring (Ex 1); decide if milk can go early (Ex 2).
3. Let them try. Students find the two flexible steps and write the whole tea order (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — boil the water (B) must come before pour the water (D).

Exercise 2 — NO: you cannot add the milk before pouring the water, because there is no tea in the cup yet; adding milk must come after pouring.

Exercise 3 — boiling (B) and getting the cup (C) can be done in either order. A working order: B, C, D, A (or C, B, D, A).

If students get stuck — and ways to adjust

Watch for: thinking every step has a fixed spot. Some steps really can swap.

Make it easier: ask just "what has to happen before you can pour?"

Make it harder: ask them to find another pair of steps that could swap.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students put step-by-step instructions in the right order to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a set of steps, change or reorder one, and explain what the change did.

You'll know it worked when

The child names the must-come-first step, the flexible pair, and writes a working order.